Path Integral Control : Theory & Practice

Goals :

- Through the derivation of the path integral control framework, gain deep understanding of its theoretical/practical advantages.
- Gain experience of path integral control algorithms, ideally on GPU environments.

Schedule :

Lecture 1 Exercise 1 Lecture 2	→ Tu	6/23
Exercise 2	→ W	6/24
Lecture 3 Exercise 3 Lecture 4	→ Th	6/25
Exercise 4	→ F	6/26.

Resources : The following lecture resources are available on GitHub:

https://github.com/tx-tanaka/kl_control_assignments

- Lecture notes
  - Ch 1 : Introduction to SOC
  - Ch 2 : Discrete-time to SOC
  - Ch 3 : Reinforcement learning
  - Ch 4 : Stochastic calculus
  - Ch 5 : Continuous-time SOC

    $\Rightarrow$ Supplementary materials

  - Ch 6 : Variational formula (Lec 1)
  - Ch 7 : Linearly solvable MDPs (Lec 2)
  - Ch 8 : Path integral control (Lec 3, 4)
- Assignments

  $\Rightarrow$ Summarizes coding exercises
- Python scripts

  $\Rightarrow$ Will be used for demos and coding assignments.

Ch 6. Variational Formula

- Appears broadly in statistics, ML, thermodynamics and information theory.
- Todorov derived the linearly-solvable MDP framework based on variational formula.
- We will derive PIC as a natural generalization of linearly solvable MDPs.

Kullback-Leibler (KL) divergence

Also known as relative entropy.
Let $Q(x)$ and $R(x)$ be two PMFs on a discrete space $\mathcal{X} = \{1, 2, \cdots, N\}$.

— ①

① is well-defined if $R(x) = 0 \Rightarrow Q(x) = 0$

R is absolutely continuous w.r.t Q, denoted by $Q \ll R$.

- $D(Q \| R) \geq 0$ for all Q and R.

  $D(Q \| R) = 0$ iff $Q = R$

  $D(Q \| R) \neq D(R \| Q)$

  Hence $D(\cdot \| \cdot)$ can be thought of as a asymmetric distance between two probability distributions.

- If $R = \text{unif}(\mathcal{X})$, then

What if $\mathcal{X}$ is a continuous space?

Let Q and R be two probability measures on $\mathcal{X}$.

$D(Q \| R) =$

- The function $\frac{dQ}{dR}(x)$ is called the Radon-Nikodym derivative. It is a function $f(x)$ satisfying

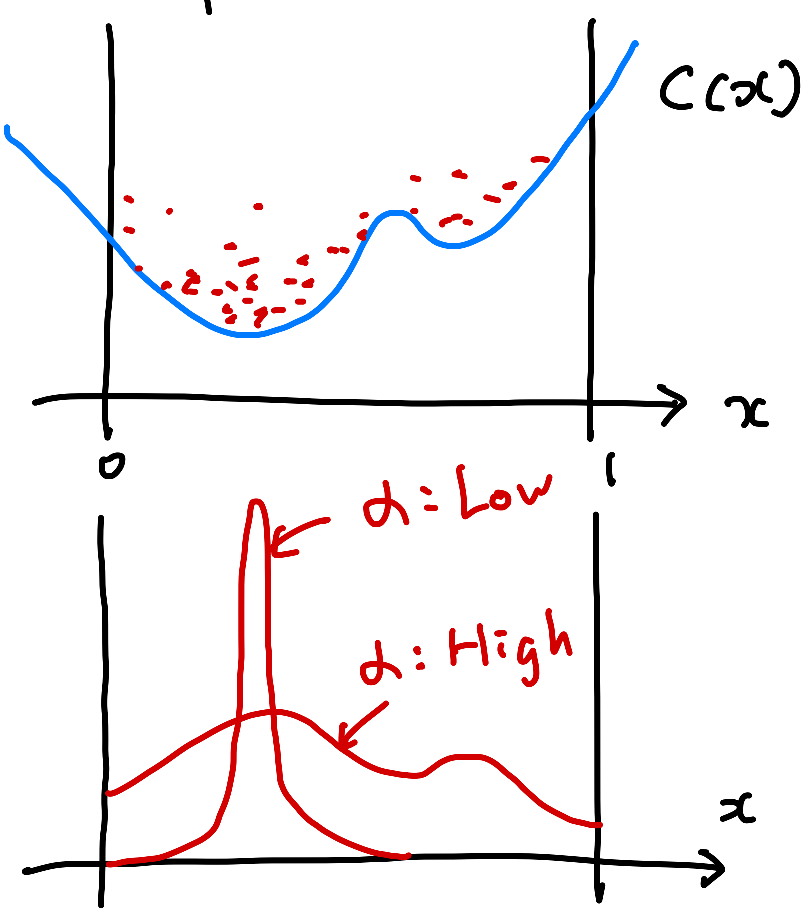
$$\int_B P(dx) = \int_B f(x)\, Q(dx)$$

for all (Borel) subset of $\mathcal{X}$.

If such a function exists, we write $Q \ll R$.

Variational Formula

Interpretation



The optimal distribution Q^* that minimizes the LHS of ② is given by

How to compute Z by Monte Carlo?

Assume we can draw iid samples $x(i)$ from R.

① Generate iid samples $x(i) \sim R$, $i = 1, 2, \cdots, N$.

② Compute the "desirability" score

$$r(i) = \exp\left(-\frac{c(x(i))}{\alpha}\right)$$

③ By the strong LLN

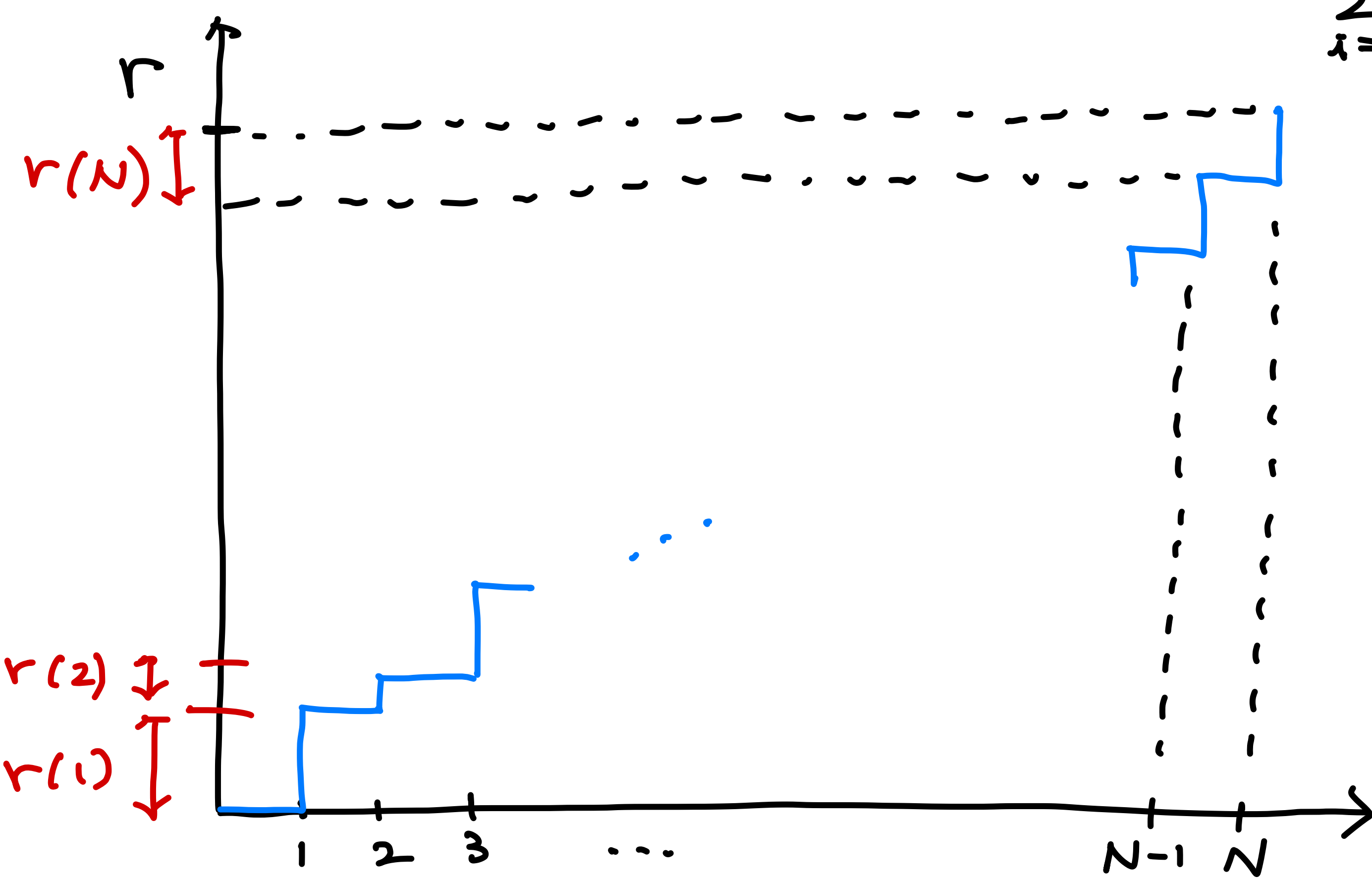
How to draw samples from Q^* ?

The algorithm below is an application of a simple "inverse transform method". See Appendix A of Krishnamurthy's textbook for other sampling methods.

① —— same ——

② —— same ——

③ Construct a staircase function $F : [0, N] \to [0, r]$, $r = \sum_{i=1}^{N} r_i$



4. Generate $d \sim \mathrm{unif}[0, r]$ randomly and pick $i \leftarrow F^{-1}(d)$.

5. Select $\tilde{X} = x(i)$

$\Rightarrow$ See Algorithm 10 of lecture note

Claim Let $\{x(i)\}_{i=1}^{N}$ be the set of samples generated by $x(i) \overset{iid}{\sim} R$. For a fixed $\{x(i)\}_{i=1}^{N}$, denote by $\Pr\{\tilde{X} \in B \mid \{x(i)\}_{i=1}^{N}\}$ the probability that $\tilde{X}$ selected by Algorithm 1 satisfies $\tilde{X} \in B$. Then, for each B,

$$\Pr\left\{\tilde{X} \in B \,\middle|\, \{x(i)\}_{i=1}^{N}\right\} \xrightarrow{a.s.} Q^*(B).$$

Proof $\Rightarrow$ See lecture note.